How to use: Fish_spots_batch_v4.py

Fits every detected spot, assigns it to soma / dendrite / axon, filters it, counts spot densities for the whole project and writes QC overlays. pipeline version 4 (October 2026).

> **Run after:** spot detection (napari/Spotiflow or `spotiflow_batch.py`), soma masks and Ilastik maps. **Run before:** every statistics and analysis script - they all read its results.

1. What you need

All scripts need Python 3 with: `pip install numpy pandas scipy scikit-image tifffile imagecodecs matplotlib openpyxl statsmodels`. `imagecodecs` is required to read the LZW-compressed Ilastik maps; without it every image fails in pass 1. Keep all scripts in ONE folder - several import each other.

Point the script at the top folder (e.g. "biological replicates"). Every image folder below it is found automatically. Folder names must contain the information the script parses:

Item	Taken from	Example
Timepoint	'div' + number in the replicate folder name	..._div27_smFISH_protease_n=3
Replicate	'n=' + number in the replicate folder name	n=3 → n3
Probe	one of PROBES in the image folder name	ProbeA prot
Condition	'prot' in the image folder name → protease, else control	ProbeA / ProbeA prot
Pixel size	calibration of each Cy5 MIP; "cm" (VisiView) → VISIVIEW_PX_UM	0.065 or 0.13 µm/px

Per image the folder must contain, all sharing the same sample ID (the text before `_C1_`, `_w1conf` or `_conf`):

File	Pattern
smFISH MIP	`...Cy5_MIP.tif`
Soma mask (manual)	`... [1].tif`
Spot coordinates	`Spots (...).csv` (napari: axis-0 = y, axis-1 = x)
Dendrite map (Ilastik)	`..._Probabilities_0.tif(f)` without "Neurite"
Neurite / filler map (Ilastik)	`...Neurite..._Probabilities_0.tif(f)`

> Images without a soma mask are ignored. `Ctrl_1` never matches `Ctrl_10`.

2. Settings (top of the script)

Setting	Default	What it does / when to change
PROBES	6 names	probe names searched in folder names; longer names first (e.g. NAME2L before NAME2)
PROB_THRESHOLD	0.35	Ilastik probability threshold for dendrite and neurite masks
DENDRITE_DILATION_UM, AXON_DILATION_UM	0.0	widen the thin masks; **0.1 µm recommended** (= exactly 0.13 µm on every camera, see technical documentation). Soma never changes
MIN_DIAMETER_NM / MAX_DIAMETER_NM	140 / 500	spot size filter (2σ)
SATURATION_LEVEL	65535	spots with a saturated pixel are rejected
MIN_SNR_BY_SETUP	5.0 ×4	SNR cutoff per camera setup (2304, 2048, 1024 Nikon, 1024 VisiView)
MIN_SNR_OVERRIDES	empty	stricter cutoff for a pair, a replicate × DIV batch, a replicate or a DIV (most specific wins; never different for control vs protease)
INTENSITY_CUTOFF	'off'	'manual' applies cutoffs you enter in intensity_cutoffs.csv
REFIT	False	True = redo all Gaussian fits (rarely needed)
QC_OVERLAYS_PER_FOLDER	3	number of QC overlays per folder; 'all' (≈8 GB, +30-60 min) or 0
QC_COLORS, QC_COUNTED_AS_RING	see script	channel colours (RGB) and ring/disc markers
N_WORKERS	CPU cores − 1	parallel processes for fitting

3. Running it

1. Open an Anaconda Prompt, `conda activate` <env>, cd to the script folder.
2. `python Fish_spots_batch_v4.py` and select the top folder.
3. Read the WARNING lines at the start (missing or ambiguous files). Read the table "SNR cutoff per imaging batch" at the start of pass 2 - it shows which cutoff each batch received.
4. First run: hours (fitting, parallel). Later runs: minutes.

What triggers what

You changed…	What happens
SNR cutoffs, size limits, intensity cutoff, QC colours	fits reused - rerun takes minutes
Spot CSVs (e.g. redetected images)	only changed images are refitted, automatically
PROB_THRESHOLD or mask dilation	masks, compartment assignment and areas rebuilt automatically; fits kept
Soma masks or Ilastik maps themselves	set REFIT = True once (or delete that folder's _pipeline_cache)

4. Outputs

File	Content
_density_results/counts_per_image.csv	**main result**: one row per image × compartment - n_spots, area_um2, density_per_um2 + timepoint, replicate, probe, condition, pair, setup
_density_results/pair_summary.csv	control vs protease per pair; density and spot-brightness ratios
_density_results/kept_spots_all_images.csv.gz	every counted spot with its measurements
_density_results/SNR_cutoff_per_pair.csv	SNR cutoff used for each pair (for the methods)
_density_results/intensity_cutoffs.csv + _per_pair.png	proposed cutoff per pair (blue dotted line); enter cutoff_manual to apply one
_density_results/Analysis_parameters_v4.txt	all settings of the run
_density_results/inventory.csv	every image found, with pixel size and setup
<image folder>/SpotCounts_v4.csv	the counts of that folder
<image folder>/QC_Overlays_v4/*_QC_v4.tif	6-channel Fiji composite (section 5)
<image folder>/_pipeline_cache/	cached fits, filtered spots (with rejection reason), areas, mask settings

5. Checking the QC overlays in Fiji

Drag a file from `QC_0verlays_v4` into Fiji; Image › Color › Channels Tool toggles the channels: C1 smFISH (grey), C2 soma outline (yellow), C3 dendrite (cyan), C4 axon (green), C5 counted spots (dark blue rings), C6 rejected spots (red circles). Adjust contrast on C1. Blue should sit on real mRNAs, red on noise and dim puncta. Judge without looking at the condition.

6. Choosing SNR cutoffs and dilation

1. Start with all SNR cutoffs at 5 and dilation 0.1 µm.
2. Open 3-5 overlays per imaging batch. If a batch shows false positives in blue, raise its cutoff (setup entry or override) and rerun (minutes). Stop when real spots start turning red.
3. Decide from the images only - never tune a batch until its densities "match" another batch.
4. Keep the final values for the whole project; `SNR_cutoff_per_pair.csv` documents them.

Troubleshooting

Message / symptom	Cause and fix
"requires the imagecodecs package"	`pip install imagecodecs`
"No image produced results in pass 1"	every image failed - read the ERROR lines above it
WARNING: n spots CSV files for <ID>	duplicate spot file (e.g. "... [1].csv") - keep one
cannot parse timepoint/replicate/probe	folder name lacks div#, n=# or a name from PROBES
WARNING: setup ... not in MIN_SNR_BY_SETUP	a new camera setup - add a line for it
image shapes differ	mask made on a cropped/rescaled image - re-export at MIP size
OMP / torch errors	not used by this script - see spotiflow_batch guide